

Review

$X \sim \mathcal{P}(\mu)$: Poisson distribution if $P\{X=n\} = \pi_n(\mu) = \frac{e^{-\mu} \mu^n}{n!}$.

$$E(X) = \text{Var}(X) = \mu.$$

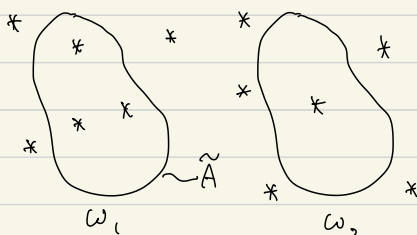
$\Pi: \Omega \rightarrow S^\infty$ random countable subset

$$A \subseteq S \quad N(A): \Omega \longrightarrow \{0, 1, 2, \dots, \infty\} \quad N(A) = \#\{\Pi \cap A\}$$

$$\omega \longmapsto \#\{\Pi(\omega) \cap A\}$$

$$\omega_1: \text{---} \underset{0}{\circ} \text{---} [\text{---} * * *]_1 \text{---} * * \text{---} \quad A = [0, 1] \quad N(A)(\omega_1) = 2$$

$$\omega_2: \text{---} \underset{0}{\circ} \text{---} [\text{---}]_1 \text{---} * * * \text{---} \quad N(A)(\omega_2) = 0$$



$$N(\tilde{A})(\omega_1) = 3$$

$$N(\tilde{A})(\omega_2) = 1$$

□ Measurability ($\mathcal{R}, \mathcal{F}, \mathbb{P}$)

$\mathcal{F}$ is a σ -field (σ -algebra) if ① $\emptyset \in \mathcal{F}$ ② $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

$$\text{③ } A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$$

If $A \in \mathcal{F}$, we say A is a measurable (m'ble) set.

A is m'ble

$$\text{Prop } \text{① } \Omega \in \mathcal{F} \quad \text{② } A_1, \dots, A_n \in \mathcal{F} \Rightarrow \bigcup_{i=1}^n A_i \in \mathcal{F} \quad \text{③ } A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcap_{i=1}^{\infty} A_i \in \mathcal{F}.$$

$$\text{Ex } \Omega = \mathbb{R}^d, \mathcal{B}(\mathbb{R}^d) = \text{Borel } \sigma\text{-algebra}$$

= smallest σ -algebra which contains all open sets.

$$A \in \mathcal{B}(\mathbb{R}^d) \Rightarrow A: \text{Borel set.}$$

Q) What does a Borel set look like?

$$(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2) \Rightarrow \mathcal{F}_1 \times \mathcal{F}_2: \text{smallest } \sigma\text{-algebra which contains } \forall A \times B, \\ A \in \mathcal{F}_1, B \in \mathcal{F}_2$$

Thm $E \in \mathcal{F}_1 \times \mathcal{F}_2 \Rightarrow E_x \in \mathcal{F}_2$ and $E^y \in \mathcal{F}_1$ $\forall x \in \Omega_1, y \in \Omega_2$.

where $E_x = \{y : (x, y) \in E\}$, $E^y = \{x : (x, y) \in E\}$.

pf) $\mathcal{A} = \{E \in \mathcal{F}_1 \times \mathcal{F}_2 : E_x \in \mathcal{F}_2\}$.

Check ① $A \times B \in \mathcal{A}$ for $\forall A \in \mathcal{F}_1, B \in \mathcal{F}_2$

② $\mathcal{A}$ is σ -algebra.

$\Rightarrow \mathcal{F}_1 \times \mathcal{F}_2 \subset \mathcal{A} \quad \therefore \forall E \in \mathcal{F}_1 \times \mathcal{F}_2, E_x \in \mathcal{F}_2$

Def A fn $f: \Omega_1 \rightarrow \Omega_2$ is m'ble if $f^{-1}(B)$ is m'ble for $\forall$ m'ble $B \subseteq Y$.

Now, fix $(\Omega, \mathcal{F}, \mathbb{P})$, (S, Σ) ^{state space}

We assume $D = \{(x, y) : x = y\} \subseteq S \times S$ is m'ble ^{Σ σ -alg.}

$\Rightarrow D_x = \{x\}$ is m'ble for $\forall x \in S$.

In this chapter, $S = \mathbb{R}^d$, $\Sigma = \mathcal{B}(\mathbb{R}^d)$ (Chapter 1, S : manifold)

II Poisson Process

$\Pi: \Omega \rightarrow S^\infty$ is a Poisson process if

(i) for any disjoint m'ble sets $A_1, A_2, \dots, A_n$ of S , the random variables $N(A_1), N(A_2), \dots, N(A_n)$ are independent.

(ii) $N(A)$ has Poisson distribution $\mathcal{P}(\mu)$ where $\mu = \mu(A)$, $0 \leq \mu \leq \infty$.

$\mu(A) = \mathbb{E}[N(A)]$.

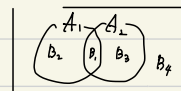
If $A_1, A_2, \dots$ disjoint, $A = \bigcup_{i=1}^{\infty} A_i$, then $N(A) = \sum_{n=1}^{\infty} N(A_n)$

Take $\mathbb{E}$ on both sides $\Rightarrow \mu(A) = \sum_{n=1}^{\infty} \mu(A_n)$. $\mu(\emptyset) = 0$.

μ : "mean measure" of Π .

Ex) How to calculate $\text{Cov}(N(A_1), N(A_2))$?

$$\text{Cov}(N(A_1), N(A_2)) = \mathbb{E}[N(A_1)N(A_2)] - \underbrace{\mathbb{E}[N(A_1)]}_{=\mu(A_1)} \underbrace{\mathbb{E}[N(A_2)]}_{=\mu(A_2)}$$



$$\begin{aligned} \mathbb{E}[N(A_1)N(A_2)] &= \mathbb{E}[(N(B_1) + N(B_2))(N(B_1) + N(B_2))] \\ &= \mathbb{E}[N(B_1)^2 + N(B_1)N(B_2) + N(B_2)N(B_1) + N(B_2)^2] \end{aligned}$$

$$\begin{aligned}
 &= \mu(B_1)^2 + \mu(B_1) + \mu(B_1)\mu(B_3) + \mu(B_1)\mu(B_2) + \mu(B_2)\mu(B_3) \\
 &= (\mu(B_1) + \mu(B_2))(\mu(B_1) + \mu(B_3)) + \mu(B_1) \\
 &= \mu(A_1)\mu(A_2) + \mu(A_1 \cap A_2)
 \end{aligned}$$

$$\therefore \text{Cov}(N(A_1), N(A_2)) = \mu(A_1 \cap A_2)$$

* Property of mean measure

Suppose that $\mu(\{x\}) = m > 0 \Rightarrow P(N(\{x\}) \geq 2) = [-e^{-m} - me^{-m}] > 0$ ($\neg$ X)

$\therefore \mu(\{x\}) = 0$ for $\forall x \in S$: "non-atomic".

(Q) μ is non-atomic $\Rightarrow \mu$ is mean measure (Section 2.5)

(Ex) $\mu(A) = \int_A \lambda(x) dx$: non-atomic. λ = rate, intensity
 $\uparrow$ positive m'ble f'n

If $\lambda(x) = \lambda$, $\mu(A) = \lambda \text{Vol}(A)$: uniform or homogeneous Poisson process

(Ex) $S = \mathbb{R}$ $\mu(t) = \begin{cases} \mu(\cdot, t] = \mathbb{E}[N(0, t)] & (t \geq 0) \\ -\mu(t, \cdot] = -\mathbb{E}[N(t, 0)] & (t < 0) \end{cases}$

$$\mu(a, b] = \mu(b) - \mu(a)$$

$\uparrow$
 Stieltjes measure

μ should be continuous ($\because$ non-atomic)

[2] Superposition Thm

Thm Let $\Pi_1, \Pi_2, \dots$ be a countable collection of independent Poisson processes on S and let Π_n have mean measure μ_n for each n .

Then $\Pi = \bigcup_{n=1}^{\infty} \Pi_n$ is a Poisson process with mean measure $\mu = \sum_{n=1}^{\infty} \mu_n$

Disjointness Lemma

Π_1, Π_2 : independent Poisson process on S , A : m'ble, $\mu_1(A), \mu_2(A) < \infty$
 $\Rightarrow \Pi_1$ and Π_2 are disjoint with probability 1 on A :

$$P(\Pi_1 \cap \Pi_2 \cap A = \emptyset) = 1.$$

Sketch of proof

$A^\dagger$: set of all finite subsets Λ of A . $\Pi_1 \cap A : p_1$ $\Pi_2 \cap A : p_2$

Define $\eta : A^\dagger \times A^\dagger \rightarrow (A \times A)^\dagger$ by $\eta(\Lambda_1, \Lambda_2) = \Lambda_1 \times \Lambda_2$: m'ble (check)

$D_A := D \cap (A \times A)$. $J = \eta^{-1}(\{\Lambda : \#(\Lambda \cap D_A) = 0\})$: m'ble (check)

$P(\pi_1 \cap \pi_2 \cap A = \emptyset) = 1 \Leftrightarrow (P_1 \times P_2)(J) = 1$. Use Fubini...

pf of Thm

$N_n(A) = \# \{ \pi_n \cap A \}$. By Lemma, π_n are disjoint on A , so $N(A) = \sum_{n=1}^{\infty} N_n(A)$.

By countable additivity Thm, $N(A) \sim \mathcal{P}(\mu)$, $\mu = \mu(A) = \sum_{n=1}^{\infty} \mu_n(A)$

$A_i \sim A_n$: disjoint. $\forall N_n(A_j)$ $n=1,2,\dots$ $j=1,2,\dots,k$ are independent

$\Rightarrow N(A_1), N(A_2), \dots, N(A_k)$ are independent. $\square$

Restriction Thm

π : Poisson process on S , mean measure μ $S_i \in S$.

Then $\pi_i = \pi \cap S_i$: Poisson process on S with mean measure $\mu_i(A) = \mu(A \cap S_i)$.

3) The Mapping Thm

Thm π : Poisson process with mean measure μ . μ : σ -finite $\left(S = \bigcup_{n=1}^{\infty} S_n, \mu(S_n) < \infty \right)$
on S . $f: S \rightarrow T$ is m'ble f'n, $\mu^*(B) = \mu(f^{-1}(B))$ is non-atomic.

Then $f(\pi)$ is a Poisson process on T with mean measure μ^* .

pf) ① $\mu(S) < \infty$, A : m'ble set.

π_1, π_2 : restrictions of π to A and A^c . $\Rightarrow \pi_1$ & π_2 are indep Poisson pro.

$\Rightarrow$ (similar argument to Lemma) $f(\pi_1)$ & $f(\pi_2)$ are disjoint with prob 1.

Let M : measure on $S \times S$ s.t. $M(C)$ is # of $(x,y) \in C$ which $x \in \pi, y \in \pi$ $f(x) = f(y)$.

Since $f(\pi_1)$ & $f(\pi_2)$ are disjoint, $M(A \times A^c) = 0$ for $\forall A$.

$\Rightarrow \forall A, B \in S$, $A \times B = \underbrace{[(A \cap B) \times (A \cap B)]}_{\text{in } A \times A^c \text{ or } B \times B^c} \cup \dots$

$\Rightarrow M(A \times B) = M((A \cap B) \times (A \cap B)) = m(A \cap B)$ ($m(A) := M(A \times A)$)

$B = S \Rightarrow m(A) = M(A \times S) \Rightarrow m$: measure

M_1 : induced measure from m on diagonal $D \subseteq S \times S$ by $d: x \mapsto (x,x)$
i.e. $M_1(E) := m(d^{-1}(E \cap D)) \Rightarrow M_1(A \times B) = m(A \cap B) = M(A \times B)$

$\Rightarrow$ By uniqueness of product measure, $M_1 = M$.

$\Rightarrow M$ is concentrated on D . $\Rightarrow \exists x,y \in M$ with $f(x) = f(y)$ with prob 1.

$$N^*(B) = \# \{f(\pi) \cap B\} = \# \{x \in \pi : f(x) \in B\} = \# \{x \in \pi : x \in f^{-1}(B)\} = N(f^{-1}(B))$$

$$\sim P(\mu(f^{-1}(B))) = P(\mu^*(B))$$

$$\textcircled{2} S = \bigcup_{n=1}^{\infty} S_n, \mu(S_n) < \infty.$$

$\pi_n := \pi \cap S_n \Rightarrow \pi_n$ are independent Poisson process with finite mean measure $\mu_n (\mu(S_n) < \infty)$

$\Rightarrow f(\pi_n)$ are indep Poisson processes with mean measures

$$\mu_n^*(B) = \mu_n(f^{-1}(B)) \text{ by } \textcircled{1}$$

$f(\pi) = f\left(\bigcup_{n=1}^{\infty} \pi_n\right) = \bigcup_{n=1}^{\infty} f(\pi_n)$ is Poisson process with

$$\mu^*(B) = \sum_{n=1}^{\infty} \mu_n^*(B) = \sum_{n=1}^{\infty} \mu(f^{-1}(B) \cap S_n) = \mu(f^{-1}(B))$$

Independence? $B_1 \sim B_n$: disjoint $\Rightarrow f^{-1}(B_1) \sim f^{-1}(B_n)$: disjoint $\square$

* Use of Thm

$\textcircled{1} \pi$: rate ftn $\lambda(x_1, x_2, \dots, x_D)$.

Take $f: \mathbb{R}^D \rightarrow \mathbb{R}^d$ to be the projection $f(x_1, \dots, x_D) = (x_1, \dots, x_d)$, $d \leq D$.

$$\mu^*(B) = \int \dots \int_{B \times \mathbb{R}^{D-d}} \lambda(x_1, \dots, x_D) dx_1 \dots dx_D = \int \dots \int_B \lambda^*(x_1, \dots, x_d) dx_1 \dots dx_d$$

$$\text{where } \lambda^*(x_1, \dots, x_d) = \int_{\mathbb{R}^{D-d}} \lambda(x_1, \dots, x_D) dx_{d+1} \dots dx_D \dots (*)$$

μ^* is non-atomic

If (*) converges, then $f(\pi)$ is a Poisson process with rate λ^* on $\mathbb{R}^d$.

$\therefore$ projection of π is again a Poisson process.

$\textcircled{2} \pi$: uniform with rate λ in the plane.

Transform to plane coordinates by $f(x, y) = ((x^2 + y^2)^{1/2}, \tan^{-1}(\frac{y}{x}))$

$$\Rightarrow \mu^*(B) = \mu(f^{-1}(B)) = \iint_{f^{-1}(B)} \lambda dx dy = \iint_B \lambda r dr d\theta \Rightarrow \lambda^*(r, \theta) = \lambda r.$$

$\textcircled{3} M(t) = \mu(0, t]$

M : conti, increasing $\mu(a, b] = M(b) - M(a)$ Take $f = M$.

$x = M(t)$, then $\mu^*((0, x]) = \mu^*((0, M(t)]) = \mu((0, t]) = M(t) = x$.

$\Rightarrow U(\Pi)$ is uniform Poisson process of rate 1.

(In ID, the uniform Poisson process is of fundamental importance)

[4] Bernoulli process

Recall X : Bernoulli distribution $X = \begin{cases} 0 & 1-p \\ 1 & p \end{cases} \quad \mathbb{E}[X] = p$.

Binomial distribution

$\hookrightarrow$ "n-times" $\mathbb{E}[X] = np$.

pick $A \Rightarrow$ get binomial distribution.

A random finite set $\Pi \subseteq S$ with $\# \Pi = n$ and

$$P(N(A_1) = n_1, \dots, N(A_k) = n_k) = \frac{n!}{n_0! n_1! \dots n_k!} p(A_0)^{n_0} p(A_1)^{n_1} \dots p(A_k)^{n_k}$$

is called a Bernoulli process. $\mathbb{E}[N(A)] = np(A)$

$$P_n(\cdot) = P(\cdot \mid N(S) = n)$$

$$A_0 = S \setminus (A_1 \cup \dots \cup A_k)$$

$$n_0 = n - \sum_{j=1}^k n_j$$

Let $A_1, \dots, A_k$ be disjoint subsets of S . Then

$$\begin{aligned} P_n(N(A_1) = n_1, \dots, N(A_k) = n_k) &= P(N(A_0) = n_0, \dots, N(A_k) = n_k) / P(N(S) = n) \\ &= \frac{\prod_{j=1}^k \frac{e^{-\mu(A_j)} \mu(A_j)^{n_j}}{n_j!}}{\frac{e^{-\mu(S)} \mu(S)^n}{n!}} \\ &= \frac{n!}{n_0! n_1! \dots n_k!} \left(\frac{\mu(A_0)}{\mu(S)} \right)^{n_0} \dots \left(\frac{\mu(A_k)}{\mu(S)} \right)^{n_k} \end{aligned}$$

$\therefore$ Conditioning $N(S) = n$ converts Poisson to Bernoulli

mean measure μ to prob measure $p(A) = \frac{\mu(A)}{\mu(S)}$

[5] The Existence Theorem

Then μ : non-atomic measure on S which can be expressed as $\mu = \sum_{n=1}^{\infty} \mu_n$, $\mu_n(S) < \infty \Rightarrow \exists$ a Poisson process on S having μ as its mean measure.

pf) WLOG suppose $\mu_n(S) > 0 \quad \forall n$.

Construct indep r.v.'s $N_n, X_{nr} (n, r \in \mathbb{N})$ s.t.

$N_n \sim P(\mu_n(S))$ and X_{nr} has distribution $p_n(\cdot) = \frac{\mu_n(\cdot)}{\mu_n(S)}$

Write $\Pi_n = \{X_{n1}, X_{n2}, \dots, X_{nN_n}\}$, $\Pi = \bigcup_{n=1}^{\infty} \Pi_n$

Write $\mu_n(A) = \# \{ \Pi_n \cap A \}$,

For disjoint $A_1, A_2, \dots, A_k$ and $A_0 = S \setminus (A_1 \cup \dots \cup A_k)$

$$P(N_n(A_1) = m_1, \dots, N_n(A_k) = m_k \mid N_n = m) = \frac{m!}{m_0! \dots m_k!} p_n(A_0)^{m_0} p_n(A_1)^{m_1} \dots p_n(A_k)^{m_k}$$

$$m_0 = m - m_1 - \dots - m_k$$

$$\Rightarrow P(N_n(A_1)=m_1, N_n(A_2)=m_2, \dots, N_n(A_k)=m_k)$$

$$= \sum_{m=\sum_{i=1}^k m_i}^{\infty} \frac{e^{-\mu(S)} \mu(S)^m}{m!} \frac{m!}{m_1! m_2! \dots m_k!} \prod_{j=1}^k p_n(A_j)^{m_j}$$

$$= \sum_{m=\sum_{i=1}^k m_i}^{\infty} \pi_{m-\sum_{j=1}^k m_j}(\mu_n(A_0)) \prod_{j=1}^k \pi_{m_j}(\mu_n(A_j)) = \prod_{j=1}^k \pi_{m_j}(\mu_n(A_j))$$

$\Rightarrow N_n(A_j)$ are indep r.v's with distribution $P(\mu_n(A_j))$

$\Rightarrow \Pi_n$ are indep Poisson processes with mean measure μ_n

$\therefore$ By Superposition Thm, Π is Poisson process with mean measure μ